

The Riemann Hypothesis via Band-Limitedness

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z -function). $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Z is real on $\mathbb{R}$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2}) \rightarrow +\infty$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, where $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, $R(t) = O(t^{-1/4})$, and $(d/dt) \arg R(t) = O(t^{-1/2})$.

Axiom 4 (Band-limited Paley–Wiener). Let $f \in L^2_{\text{loc}}(\mathbb{R})$. Define the windowed transform $K_T(\mu) := (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If there is $\tau \geq 0$ and a constant M with $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T , and $\lim_{T \rightarrow \infty} |K_T(\mu)|^2/T = 0$ for every $|\mu| > \tau$, then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with $|f(z)| \leq \sqrt{2\pi M} e^{\tau|\Im z|}$.

Axiom 5 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type $\leq \tau$, real on $\mathbb{R}$, and $F|_{\mathbb{R}} \in L^2(\mathbb{R})$ with its Fourier transform supported in $[-2\tau, 2\tau]$, then every zero of F in $\mathbb{C}$ is real.

2 Definitions

Definition 1. $\Theta(t) := \vartheta(t) + ct$, $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

3 Lemmas

Lemma 4 (Warp diffeomorphism). $c_\star \in (0, \infty)$; for $c > c_\star$, $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$.

Lemma 5 (Factorization). $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ for every $t \in \mathbb{R}$.

Lemma 6 (Frequency ratio). Set $\omega_n(t) := (\vartheta'(t) - \log n)/(\vartheta'(t) + c)$. For $n \geq 1$ and $t \geq 2\pi n^2$, $\omega_n(t) \in [0, 1)$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

4 Main Results

Theorem 7 (Uniform L^2 bound on $K[T]$). *There is $M > 0$ with $|K[T](\mu)|^2 \leq M \Theta(T)$ for every $T \geq 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz gives $|K[T](\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du$. Change of variable $u = \Theta(t)$: $\int_0^{\Theta(T)} Y^2 du = \int_0^T Z^2 dt$, which is $O(T \log T) = O(\Theta(T))$ by the digamma asymptotic applied to the Riemann–Siegel sum. Thus $|K[T](\mu)|^2 / \Theta(T)$ is bounded. $\square$

Theorem 8 (Band-limit). *For $|\mu| > 1$, $\lim_{T \rightarrow \infty} |K[T](\mu)|^2 / \Theta(T) = 0$.*

Proof. Insert Riemann–Siegel into $K[T](\mu)$ and substitute $u = \Theta(t)$. The (n, σ) -mode contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i(\sigma\vartheta(t) - t \log n - \mu\Theta(t))} dt.$$

The phase derivative is $\sigma\vartheta'(t) - \sigma \log n - \mu\Theta'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$. By the frequency-ratio lemma $\omega_n(t) \in [0, 1)$, so $|\sigma\omega_n(t) - \mu| \geq |\mu| - 1 =: \lambda > 0$, and by the warp-diffeomorphism lemma $\Theta'(t) \geq c - c_\star > 0$; the phase slope is thus bounded below by $\lambda(c - c_\star) > 0$ uniformly in n, t . Write the integrand as $G_{n,\sigma}(t) e^{iP(t)}$ with $G_{n,\sigma}(t) = n^{-1/2} \sqrt{\Theta'(t)}$, $P'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$; one integration by parts gives

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{2n^{-1/2} \sqrt{\Theta'(T)}}{\lambda(c - c_\star)} + \int_{2\pi n^2}^T \left| \frac{d}{dt} \frac{G_{n,\sigma}}{P'} \right| dt.$$

The integrand of the second term is $O(n^{-1/2} \Theta'(t)^{-1/2} \vartheta''(t))$ since $\vartheta''(t) = (2t)^{-1} + O(t^{-3}) > 0$, giving total mass $O(n^{-1/2} / \sqrt{\log T})$ (the $\Theta'(t)^{-1/2} \asymp (\log t)^{-1/2}$ factor against $\vartheta''(t) \asymp 1/t$ integrates to $O((\log T)^{-1/2})$). A second integration by parts against P' , using $(G_{n,\sigma}/P')' \asymp n^{-1/2} \Theta'(t)^{-1/2} \vartheta''(t) / \lambda$ and $P'' = \vartheta''(t)(\sigma - \mu) + \sigma\vartheta''(t) \cdot O(1) = O(\vartheta''(t))$, extracts an additional factor $1/(\lambda(c - c_\star)\Theta'(T)) = O(1/\log T)$ at the endpoint and an additional $1/\log T$ on the integrated remainder, giving

$$|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2} / (\log T)^{3/2}).$$

The remainder contribution $\int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt$ has phase slope $-\mu\Theta'(t) + O(t^{-1/2})$, bounded below in modulus by $|\mu|(c - c_\star)/2$ for t large by the warp-diffeomorphism lemma; one integration by parts gives $O(T^{-1/4})$.

Summing over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma \in \{\pm 1\}$:

$$|K[T](\mu)| = O\left(\sum_{n=1}^{N(T)} n^{-1/2} / (\log T)^{3/2}\right) + O(T^{-1/4}) = O(T^{1/4} / (\log T)^{3/2}),$$

so $2\pi|K[T](\mu)|^2 / \Theta(T) = O(T^{-1/2} / (\log T)^4) \rightarrow 0$, uniformly on compact subsets of $\{|\mu| > 1\}$. $\square$

Corollary 9 (Three-domain convergence of the periodogram). *Let $P_T(d\mu) := 2\pi|K[T](\mu)|^2 / \Theta(T) d\mu$. Then as $T \rightarrow \infty$:*

- (i) *On $\{|\mu| > 1\}$: P_T has density tending to 0 uniformly on compacta at rate $O(T^{-1/2} / (\log T)^4)$.*
- (ii) *On $(-1, 1)$: $P_T \rightarrow P_\infty|_{(-1,1)}$ in the weak- $\ast$ sense on compact subsets, with the phase-slope bound $|\sigma\omega_n(\Theta^{-1}(u)) - \mu| \geq 1 - |\mu| > 0$ non-uniform as $|\mu| \uparrow 1$.*
- (iii) *On $\{|\mu| = 1\}$: $\sigma\omega_n(t) \rightarrow \sigma$ is asymptotic, not attained, for every finite n, t ; mass accumulates at $\mu = \pm 1$ from inside as $T \rightarrow \infty$, producing the endpoint atoms of P_∞ .*

Fubini on $A \times [0, T]$ holds for every bounded Borel $A \subseteq \mathbb{R}$ and every finite T : by Cauchy–Schwarz, $\int_0^T |\zeta(\frac{1}{2} + it)| \sqrt{\Theta'(t)} dt = O(T \log T)$, so the double integral is bounded by $|A| \cdot O(T \log T) < \infty$.

Proof. Item (i): the band-limit theorem and its rate above. Item (ii): the phase slope is $\Theta'(t)(\sigma\omega_n(t) - \mu)$ and $\Theta'(t) \geq c - c_\star > 0$, with $\sigma\omega_n(t) - \mu \in [-\mu - 1, 1 - \mu]$ of sign determined by $|\mu| < 1$ but with modulus $\geq 1 - |\mu|$ only after taking the supremum over n ; the IBP constant $1/(\lambda(c - c_\star))$ with $\lambda = 1 - |\mu|$ blows up as $|\mu| \uparrow 1$, removing the uniform rate. Item (iii): the frequency-ratio lemma gives $\omega_n(t) < 1$ for every (n, t) with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$; the same lemma with $\sigma = -1$ gives $\sigma\omega_n(t) \downarrow -1$. The Fubini bound is Cauchy–Schwarz against $\int_0^T \Theta'(t) dt = \Theta(T) = O(T \log T)$ and the second-moment asymptotic $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt \sim T \log T$. $\square$

Theorem 10 (Entirety and exponential type). *Y extends uniquely to an entire function on $\mathbb{C}$ of exponential type ≤ 1 .*

Proof. The uniform bound theorem with M independent of T and the band-limit theorem give the two hypotheses of the Paley–Wiener axiom with $\tau = 1$, applied to $f = Y$ on $[0, \Theta(T)]$. $\square$

Theorem 11 (Reality of zeros of Y). *Every zero of Y in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Y is entire of exponential type ≤ 1 by the preceding theorem. Y is real on $\mathbb{R}$: by the factorization lemma and $\sqrt{\Theta'(t)} > 0$, $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$, and Z is real on $\mathbb{R}$ by the Hardy Z -function axiom. The windowed-transform support in the band-limit theorem, evaluated through the Paley–Wiener axiom, places the Fourier transform of $Y|_{\mathbb{R}}$ in $[-1, 1] \subseteq [-2, 2]$. The Akhiezer–Laguerre–Pólya axiom with $\tau = 1$ applies. $\square$

Theorem 12 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re \rho = \frac{1}{2}$.*

Proof. By the Hardy Z -function axiom, $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$. By the factorization lemma, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$, so $Z(t) = 0 \iff Y(\Theta(t)) = 0$. By the reality theorem, every zero of Y in $\mathbb{C}$ is real. Therefore every nontrivial zero of ζ lies on $\frac{1}{2} + i\mathbb{R}$. $\square$